

Zhuoyuan Li

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Institute for Functional Intelligent Materials
National University of Singapore

Education

2020–2025 **Ph.D.**, School of Mathematical Sciences, Peking University, Beijing, China
co-supervisors: Prof. Pingwen Zhang¹, Prof. Bin Dong²
2016–2020 **B.Sc.**, School of Mathematics, Sichuan University, Chengdu, China

Working Experience

2025–present Research Fellow Institute for Functional Intelligent Materials
National University of Singapore, Singapore

Research Interests

- Deep learning for multiscale modeling
- Numerical weather prediction and data assimilation
- Inverse problems and uncertainty quantification

Publications

🔍 [Google Scholar](#)  [ORCID](#)

† → Equal contribution; * → Corresponding author(s)

Journal Articles

- J1. Wang[†], R., Jiang[†], J., **Zhuoyuan Li**, Liu, T., Wang, Y., Xie*, M. & Piatkevich*, K. D. Fluorescent protein ticker tape (FPTT): Multiplexed recording of transcriptional dynamics in living cells and in vivo. *Science Advances* **12**, eaef9406. <https://www.science.org/doi/abs/10.1126/sciadv.aef9406> (2026).
- J2. **Zhuoyuan Li**, Dong*, B. & Zhang*, P. State-observation augmented diffusion model for nonlinear assimilation with unknown dynamics. *Journal of Computational Physics* **539**, 114240. ISSN: 0021-9991. <https://doi.org/10.1016/j.jcp.2025.114240> (2025).
- J3. **Zhuoyuan Li***, Dong, B. & Zhang, P. Latent assimilation with implicit neural representations for unknown dynamics. *Journal of Computational Physics* **506**, 112953. ISSN: 0021-9991. <https://doi.org/10.1016/j.jcp.2024.112953> (2024).

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Peer-reviewed Conference Proceedings

- C1. Yang[†], P., Feng[†], Y., Chen[†], Z., Wu, Y. & **Zhuoyuan Li**^{*}. *Spend Wisely: Maximizing Post-Training Gains in Iterative Synthetic Data Bootstrapping in Advances in Neural Information Processing Systems* **38** (2025), 158602–158639. <https://doi.org/10.52202/085713-5299>.

Preprints

- P1. **Zhuoyuan Li**^{*,†}, Zhao[†], Y. & Li, M. Ensemble Controlled-Flow Filtering for Implicit Data Assimilation. *arXiv preprint arXiv:2607.12975*. under review. <https://arxiv.org/abs/2607.12975> (2026).
- P2. **Zhuoyuan Li**[†], Zhu[†], A. & Li^{*}, Q. Hypothesis-driven construction of mesoscopic dynamics. *arXiv preprint arXiv:2605.16211*. under review. <https://arxiv.org/abs/2605.16211> (2026).
- P3. Li[†], L., **Zhuoyuan Li**[†], Li[†], S., Zhan, K., Gao^{*}, H., Chen^{*}, C. & Yang^{*}, L. In-context modeling as a retrain-free paradigm for foundation models in computational science. *arXiv preprint arXiv:2604.23098*. under review. <https://arxiv.org/abs/2604.23098> (2026).
- P4. Huang[†], X., **Zhuoyuan Li**[†], Liu, H., Wang, Z., Zhou, H., Dong^{*}, B. & Hua, B. Learning to simulate partially known spatio-temporal dynamics with trainable difference operators. *arXiv preprint arXiv:2307.14395*. <https://arxiv.org/abs/2307.14395> (2023).

Talks and posters

EASIAM 2026 CONTRIBUTED TALK <i>Hypothesis-driven construction of mesoscopic dynamics</i>	Aug. 23–27, 2026
AI4X 2026 CONTRIBUTED TALK <i>Learning non-equilibrium mesoscopic dynamics with Onsager principle</i>	Jun. 15–19, 2026
CSIAM 2024 OUTSTANDING POSTER <i>Latent assimilation with implicit neural representations for unknown dynamics</i>	Oct. 24–27, 2024

Teaching

Peking University

Advanced Algebra (II) ASSISTANT INSTRUCTOR	Spring 2023
Advanced Algebra (I) ASSISTANT INSTRUCTOR	Fall 2022
Advanced Algebra (II) ASSISTANT INSTRUCTOR	Spring 2022
Advanced Mathematics (C) TEACHING ASSISTANT ³	Fall 2020

Please see my homepage for more details.

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³without teaching tasks